

Jayanth Vennamreddy

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

- Concentration: Computing Systems

Atlanta, GA

Expected May 2027

Bachelor of Science in Computer Science | GPA: 3.79

August 2023 - May 2026

- Threads: Artificial Intelligence and Information Internetworks

EXPERIENCE

Google Summer of Code 2026 Contributor, Distributed Filesystems

May 2026 – Aug 2026

Apache Software Foundation

Remote

- Developed a **Rust filesystem interface** for AI-for-science datasets, exposing ATLAS metadata for **1,900+ proteins** and supporting access through materialized directories, **FUSE**, and **NFS**
- Implemented **dataset-agnostic metadata registry** with **TSV/CSV connectors** for ATLAS, mdCATH, GPCRmd, AlphaFold DB, and DANDI without modifying filesystem core logic

Software Engineer Intern, HPC Resource Allocation Systems

May 2026 – Jul 2026

Georgia Institute of Technology — Open Source Programs Office

Atlanta, GA

- Built a **research impact pipeline** that linked HPC compute allocations to publications using **Semantic Scholar**, **Crossref**, and **ORCID APIs**, improving visibility into compute usage and scientific outcomes
- Developed **Go REST APIs** and **React/TypeScript** admin workflows for researcher identity, account linking, RBAC, and allocation visibility, replacing mock/local UI state with live backend APIs

Machine Learning Research Engineer, Protein Modeling

Jan 2025 – May 2026

Georgia Institute of Technology (NSF NAIRR Pilot)

Atlanta, GA

- Co-lead author on **PEARC'26 ACM** paper on modular protein-structure prediction workflows, building the **ESMFold backend** for VizFold with hook-based representation extraction and reproducible HPC deployment
- Engineered embedding extraction pipelines for the 48-layer **OpenFold Evoformer**, enabling downstream ML analysis and visualization across **10K+ protein structures** on the ICE HPC cluster using **Slurm**

Open Source Systems Contributor, Distributed Databases

Jan 2026 – Present

Open Source Contributions

- Resolved **ClickHouse** query-correctness bugs involving `arrayJoin()` with JOIN and WHERE clauses, eliminating double evaluation across analyzer and planner stages
- Fixed cross-platform runtime failures, async worker lifecycle bugs, and **command-injection vulnerabilities** across open-source systems projects

PROJECTS

dbscope | Rust, PostgreSQL, MySQL, SQLite, ClickHouse

Jan 2026 – Present

- Built and maintained a database schema analysis **CLI** modeling foreign-key dependencies as a graph to compute risk and blast-radius metrics across 4 database engines
- Shipped migration simulation (**plan/preview**), schema diffing, linting (20+ rules), dependency analysis, and an **MCP** server; published to **crates.io** with cross-compiled binaries and **Docker** image via GitHub Actions **CI/CD**

ILU | Next.js, TypeScript, Prisma, PostgreSQL

Nov 2025 – Present

- Built and deployed a housing transparency platform for Georgia Tech students with a categorical review system (noise, safety, internet, cleanliness) and three-tier anonymity model, acquiring **100+** organic student reviews

KubeQueue | Go, Kubernetes, PostgreSQL, Docker, React, TypeScript

Jul 2026 – Present

- Built a Kubernetes-based batch job orchestrator for scientific workflows with a Go API, PostgreSQL metadata, React/TypeScript dashboard, and support for job submission, status tracking, logs, retries, and resource requests

TECHNICAL SKILLS

Languages: Rust, Go, Python, Java, C++, C, TypeScript/JavaScript, SQL

Infrastructure: Linux, PostgreSQL, ClickHouse, MySQL, SQLite, Kubernetes, Docker, FUSE, Slurm, GitHub Actions

ML & Data: PyTorch, scikit-learn, NumPy, Pandas

Frameworks: React, Next.js, Prisma, FastAPI, Flask